15. Probabilistic Context-Free Grammar (PCFG) Parsing

Aim: To implement probabilistic context-free grammar parsing using Python.

Procedure:

1. Import NLTK.
2. Define grammar rules with probabilities.
3. Create a probabilistic parser.
4. Parse the input sentence.
5. Display the parse tree and its probability.

Code:

```
import nltk
grammar = nltk.PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [1.0]
VP -> V [1.0]
Det -> 'the' [1.0]
N -> 'boy' [1.0]
V -> 'runs' [1.0]""")
parser = nltk.ViterbiParser(grammar)
sentence = "the boy runs".split()
for tree in parser.parse(sentence):
    print(tree)
    print("Probability:", tree.prob())
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/15.py"
(S (NP (Det the) (N boy)) (VP (V runs))) (p=1)
Probability: 1.0
```

Result: The sentence was successfully parsed using a probabilistic context-free grammar.